Yiyang Chen

chen.yiyang@wustl.edu yiyangc1999.github.io Google Scholar 0009-0004-6041-1298 GitHub

Education

Washington University in St. Louis St. Louis, MO, USA
Ph.D. in *Imaging Science* 2021 – present
M.S. in *Electrical Engineering* (en route to Ph.D.) 2025

- **Dissertation**: Multi-dimensional Single Molecule Nanoscopy to Elucidate the Structure and Conformational Dynamics of Cell Membrane Proteins
- **Advisor**: Matthew Lew
- **Dissertation Committee**: Matthew Lew, Joseph O'Sullivan, David Piston, Michael Vahey, Christy Landes

Nankai University Tianjin, China
B.S. (Honors) in *Physics*, Poling Class of Physics 2021

- **Thesis**: The Study of Deformability of Human Erythrocyte Based on Microfluidics
- **Advisor**: Leiting Pan

Publications

*equal contribution

Peer-Reviewed Publications

[1] **Y. Chen***, Y. Qiu*, M. D. Lew. Single Fluorogens and Orientation-Localization Microscopy for Quantifying Chemical and Biomolecular Dynamics at the Nanoscale. *Acc. Chem. Res.* **59**, 2693–2708 (2026). [Review] DOI:10.1021/acs.accounts.6c00353

[2] E. Gillett, S. Chatterjee*, J. Chatterjee*, N. Kovalenko, C. Xu, D. Fan, **Y. Chen**, Y. Qiu, J. Miao, V. Nelavoy, M. D. Lew, M. P. Backlund, C. F. Landes. Fused deep learning enables 6D single-molecule localization in polarization-resolved microscopy. *Methods Appl. Fluoresc.* **14**, 035006 (2026). DOI:10.1088/2050-6120/ae8eff

[3] **Y. Chen**, Y. Qiu, M. D. Lew. Resolving the Orientations of and Angular Separation Between a Pair of Dipole Emitters. *Phys. Rev. Lett.* **134**, 093805 (2025). DOI:10.1103/PhysRevLett.134.093805

Preprints & Manuscripts in Preparation

[1] X. Yan, S. Zedlitz, X. Chen, Y.-C. Tsai, Y. Ma, M. W. Chen, L. Li, D. Kuster, J. Nijssen, A. Sergeeva, **Y. Chen**, C. Welch, W. Yu, Z. Li, M. Messina, N. Karedla, J. Savage, M. D. Lew, Y. Okuno, S. Alberti, K. Pombo-García, J.-Y. Youn, Y. Dai, A. A. Hyman. Redox surveillance of stress granules prevents TDP-43 aggregation. *bioRxiv* (2026). DOI:10.64898/2026.10.16.759411. (Under review at *Science*)

[2] X. Li*, **Y. Chen***, Y. Tan, T. Wu, Y. Liu, Y. Zhang, M. D. Vahey, C. Fernandez-Granda, Q. Qu, M. D. Lew. DeepSMOLM3D: Deep Learning-Based Single-Molecule Imaging Reveals Distinct Fluorogenic Probe Orientations upon Membrane Binding. *Manuscript in preparation*.

Conference Proceedings

[1] **Y. Chen**, Y. Qiu, M. D. Lew. Combining Excitation and Emission Modulation Resolves the Angular Separation between a Pair of Dipole Emitters. In *Optica Biophotonics Congress 2025*, Technical Digest Series (Optica Publishing Group, 2025), paper NM1C.4.

Research Experience

Single-Molecule Orientation-Localization Microscopy: Fundamental Limits, Computational Methods, and Biophysical Applications 2022 – present
Graduate Research Assistant | Advisor: Matthew Lew | Preston M. Green Department of Electrical & Systems Engineering, Washington University in St. Louis

Theory & Fundamental Limits

- Established the fundamental limit on distinguishing two spatially overlapping rotating dipole emitters: identified the degeneracy that makes them indistinguishable, and proved that resolving it requires fourth-order orientation moments.
- Developed a combined excitation-polarization and emission-DSF modulation scheme that improves angular-separation precision by 200–400% and centroid-orientation precision by 50% over state-of-the-art methods.
- Currently extending the approach to direct fourth-order-moment measurement, enlarging the accessible parameter space for higher-order orientational dynamics.

Optical Instrumentation

- Recommissioned a dual-polarization 4f microscope following a spatial light modulator (SLM) upgrade: realigned the emission path, performed full characterization of the replacement SLM including look-up-table calibration, and re-sampled and re-optimized established dipole-spread-function (DSF) designs for the new modulator's parameters.

Computational Imaging Methods

- Designed dipole-spread functions optimized for targeted orientation regimes by maximizing Fisher information with convex optimization.
- Built the physics-informed simulation pipeline generating training and validation data for DeepSMOLM3D, modeling the full 6D measurement (3D position + 3D orientation).
- Co-developed the DeepSMOLM3D reconstruction algorithm, contributing to network architecture design and implementing position estimation.
- Benchmarked DeepSMOLM3D against RoSEO3D, the iterative maximum-likelihood standard, across SNR and emitter-density regimes in simulation and on cell-membrane data: comparable or better accuracy and precision with $\sim 10\times$ faster reconstruction.

Biophysical Applications

- Imaged supported lipid bilayers and cell membranes by single-molecule PAINT with membrane-sensitive fluorogens, resolving distinct binding dynamics among probes across membrane structures.
- Evaluated redox chemistry inside biomolecular condensates by single-molecule imaging of hydroxyphenyl fluorescein (HPF), discovered interface-dependent redox activity in G3BP1 stress-granule condensates.
- Quantifying how redox activity in resilin-like polypeptide condensates evolves over aging, by single-molecule HPF imaging across a 24-hour time course. *Manuscript in preparation*.
- Developing protein-tag-based labeling strategies to measure membrane-protein orientation and rotational dynamics in live cells.

Study of Human Red Blood Cell (hRBC) Membrane Characteristics Based on Microfluidics and Single-Molecule Localization Microscopy 2019 – 2021

Undergraduate Research Assistant | Advisor: Leiting Pan | School of Physics, Nankai University
Supported by the *China National University Student Innovation & Entrepreneurship Development Program*

- Analyzed the actin-spectrin network and quantified CD47 diffusivity on human red blood cell (hRBC) membranes by single-molecule localization microscopy (SMLM).
- Quantified age-related hRBC deformability from transit speeds through a microfluidic channel, and designed and simulated a microfluidic ratchet chip (COMSOL Multiphysics) to sort cells by deformability.

Immunofluorescent Biomarker for Zebrafish Somitogenesis and Single Cell Oscillation July 2019 – Sept 2019

Summer Visiting Student | Advisor: Qiong Yang | Department of Biophysics, University of Michigan

- Developed immunofluorescent biomarkers targeting signaling pathways (*Ntla* and *Tbx16*) for studying zebrafish somitogenesis and segmentation clock dynamics. Identified and characterized oscillating cells with different phenotypes in zebrafish embryos and cell dispersal systems.

Conference Presentations

[1] **"DeepSMOLM3D: Fast and Accurate 6D Single-Molecule Imaging via Deep Learning"**, poster. *7th Midwest Single Molecule Workshop*, University of Illinois Urbana-Champaign, Urbana, IL, July 2026

[2] **"Combining Excitation and Emission Modulation Resolves the Angular Separation Between a Pair of Dipole Emitters"**, oral. *Optica Biophotonics Congress: Optics in the Life Sciences*, Coronado, CA, April 2025

[3] **"Resolving the Orientations of and Separation between an Overlapping Pair of Dipole Emitters"**, poster. *Gordon Research Conferences: Single Molecule Approaches to Biology*, Newry, ME, July 2024

[4] **"Immunofluorescent Biomarkers for Distinguishing Cell Phenotypes in Zebrafish Somitogenesis and Autonomous Cellular Oscillators"**, poster. *APS March Meeting 2020*, Denver, CO, March 2020

Honors and Awards

Second-Place Poster Prize, Spectra (WashU Optica & SPIE Student Chapter) Student-Led Conference 2025

Excellence Award, China National University Student Innovation & Entrepreneurship Development Program 2021

First Prize, Nankai Physics Tournament 2018

Poling Scholarship, Nankai University 2017

Skills

Programming: Python, MATLAB, Wolfram Mathematica

Computational Methods: inverse problems, Fisher information analysis, maximum-likelihood estimation, convex optimization, deep learning for microscopy

Laboratory

- Optical Lab: fluorescence microscopy, 4f systems, polarization-resolved imaging, SLM calibration and phase modulation, optical alignment and calibration.
- Wet Lab: cell culture; immunolabeling; sample preparation; supported-lipid bilayer extrusion and coating.

Software: ImageJ, COMSOL Multiphysics, Origin, PyMOL, Zemax OpticStudio

Languages: Chinese (native), English (full professional proficiency), French (elementary proficiency)

Service & Leadership

Peer Review

Co-reviewer with M. D. Lew: *ACS Nano*; *Angewandte Chemie International Edition*; *APL Photonics*; *The Journal of Chemical Physics*

Leadership

Spectra, Optica & SPIE joint student chapter at *Washington University in St. Louis*
Co-president (Optica liaison) 2023 – 2024
Vice president (Imaging Science Pathway liaison) 2022 – 2023

Optica student chapter at *School of Physics, Nankai University*
Student officer 2018 – 2020

Additional Training

Medical Physics Summer School at *Duke Kunshan University* 2020

The Physics of Life Summer School (virtual) at the *Center for the Physics of Biological Function, Princeton University* 2020

Teaching and Mentoring

Research Mentoring

Cynthia Cheng, undergraduate researcher, Lew Lab

- *Toward Smart, Adaptive Single-Molecule Orientation Microscopy* | supported by the WashU Summer Undergraduate Research Guided Experience (SURGE) program.

Teaching

Fundamentals and Applications of Modern Optical Imaging Spring 2024
Washington University in St. Louis
Assistant Instructor | Course Instructor: Matthew Lew

College Physics Fall 2019
Nankai University
Undergraduate Teaching Assistant | Course Instructor: Jianghong Yao